Modeling Future Cases of Chronic Diseases in the U.S. by State

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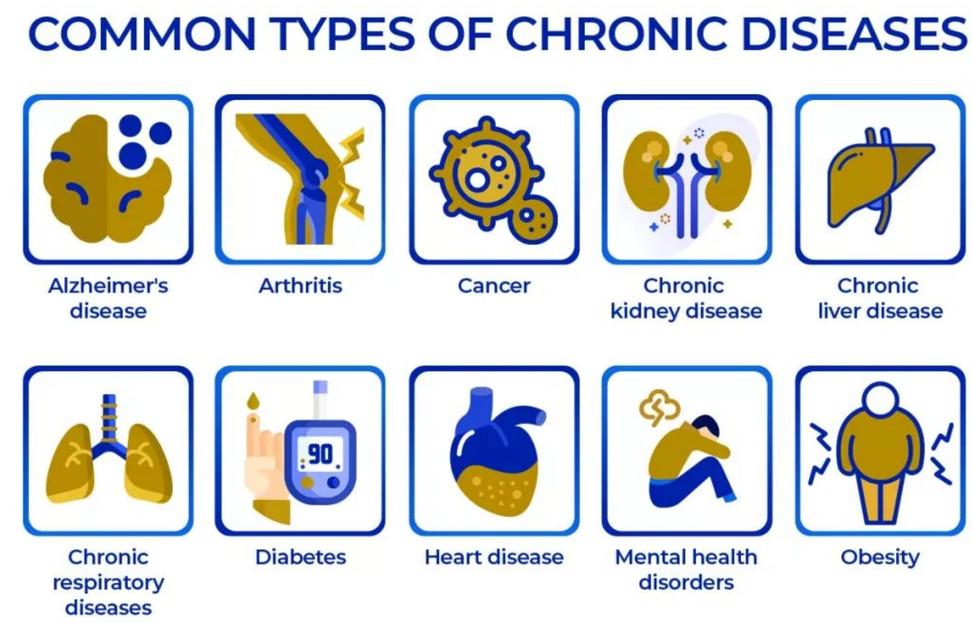
Summary

- Chronic diseases are conditions that last one year or longer and require ongoing medical attention and/or limit activities in daily living
- They are a major public health concern in the United States; in 2018, the National Health Interview Survey estimated that 129 million Americans had at least one major chronic disease
- Our study examined demographic and geographic features that predicted chronic disease mortality risks
- By implementing our study, we aimed to extract meaningful insights that could better inform patients, healthcare providers, and state health departments about chronic disease risk



Methodology

Key Features

- Tableau map dashboard with interactive filtering options
- Five chronic diseases: cancer, diabetes, chronic kidney disease, chronic obstructive pulmonary disease, and cardiovascular disease
- Forecasted future Risk Scores through ML algorithms
- Utilized state-level Medicaid expenditures–a critical dimension often overlooked

Data

Source

- CDC U.S. Chronic Disease Indicators CDI 2023 Release
- U.S. Centers for Medicare & Medicaid Services Health Expenditures by State of Residence, 1991-2020
- U.S. Census Bureau Median Household Income by State

Size

- Largest file: 359MB raw data in CSV and 668,502 observations from 2008-2021
- Others: 190KB and 105KB

Model

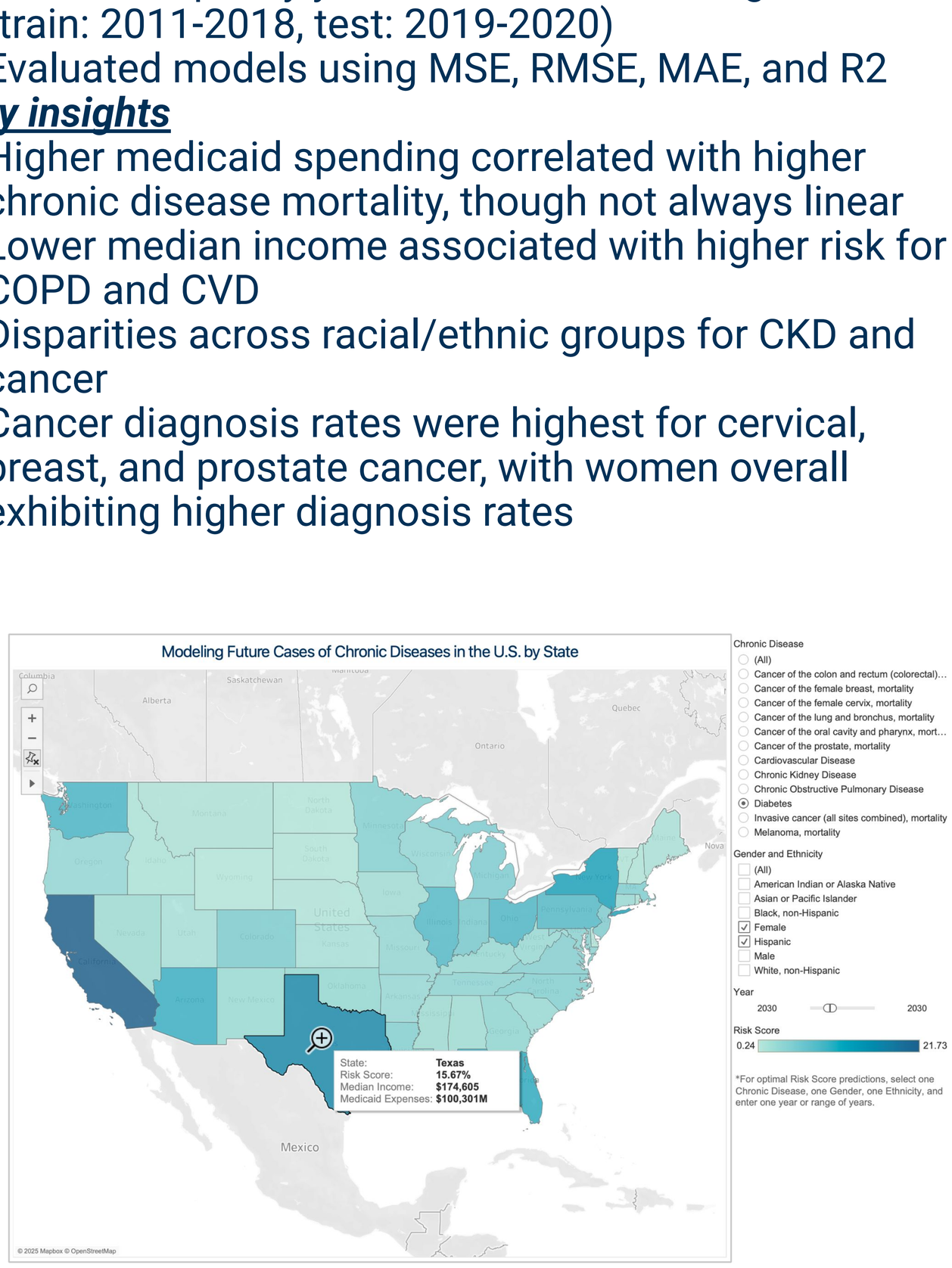
- Modeled the five chronic diseases individually
- Identified best-performing machine learning model per disease
- Cancer, Diabetes: XGBoost Regression
- CVD, CKD, COPD: Random Forest
- Forecasted disease mortality through year 2050 using historical data and demographic/economic inputs

Test and Evaluation

- Train/test split by year to avoid data leakage (train: 2011-2018, test: 2019-2020)
- Evaluated models using MSE, RMSE, MAE, and R2

Key insights

- Higher medicaid spending correlated with higher chronic disease mortality, though not always linear
- Lower median income associated with higher risk for COPD and CVD
- Disparities across racial/ethnic groups for CKD and cancer
- Cancer diagnosis rates were highest for cervical, breast, and prostate cancer, with women overall exhibiting higher diagnosis rates

Visualization

- Developed map visualization using Tableau
- Predictions dataset from ML algorithms loaded into Tableau
- Map shows future Risk Scores based on interactive, filterable options such as chronic disease, gender, ethnicity, and year
- Tooltip shows the state name, Risk Score, median income, and Medicaid expenditures